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On Rajecki, D.W.; Lamb, M.E.; and Obmascher, P. (1978) Toward a general theory of infantile attachment: a comparative review of aspects of the social bond. BBS 1:417–464.

Abstract: This critical appraisal of contemporary interpretations in the area of infantile attachment begins with an outline of the principal features of the Bowlby-Ainsworth ethological theory, the instrumental/operant learning theory of Gewirtz, and Hoffman's classical conditioning model. Some attention is also given to Cairns's contiguity learning analysis and the Hoffman-Solomon opponent-process model. Discussion of these theories is followed by a review of representative data from infants at four phyletic levels (precocial birds, dogs, monkeys, and human beings), with an emphasis on three aspects of social bonding: (a) the formation and persistence of social ties in the infant under conditions of maltreatment, (b) the role of the attachment object in the adjustment of the infant to the broader environment (the so-called secure base effect), and (c) the infant's reaction to involuntary separation from the attachment object.

An attempt is made to judge how well each of the interpretations accounts for all or part of the data, with the conclusion that current theories do not accord completely with documented attachment phenomena. The following criticisms are highlighted: Ethological theory emphasizes that infants' behavior systems have been shaped by the ordinarily expectable environment and depend on that environment for their functioning, yet infants of many species form bonds to objects not typical in any species' environment, or even to sources of maltreatment. Learning theory is faulted for making predictions contradicted by the maltreatment data and for a lack of formal mechanisms to account for the secure base and separation effects. The contiguity analysis is criticized for its inability to account for the emergence of certain response patterns during separation, and the opponent-process model is called into question because of its failure to fit important affective dynamics of social separation (a central focus of this theory). Recommendations for future theories of attachment are offered.

by Gerard P. Baerends

Department of Zoology, University of Groningen, Haren, Netherlands

Programmed development

Although I greatly appreciate the successful effort of Rajecki et al. (1978) to produce a concise and useful survey of our present, experimentally-based knowledge concerning the causation of infantile social bonding, I am afraid that the underlying aim of the review: "to evaluate *alternative* explanations of the psychology of infantile attachment," and, in particular, the philosophy with which this is done, is unfortunate. The conviction that one shall have to make a choice among one of these or possibly other comparable theories and hypotheses to approach the best causal explanation for infantile social bonding may not only turn out to be a red herring, but it will in addition seriously impede further research. One reason for this is in fact mentioned by the authors: the differences one may expect between species. The reason I wish to stress here is that it is unjustified to consider infantile social bonding – even in one species – as a unitary process, for which only one of the explanatory principles is likely to hold. For the three aspects given particular action – maltreatment effects, secure basis effects, and separation effects – this seems extremely unlikely.

It seems to me that the authors have insufficiently appreciated that the ethologically-inspired theory of Bowlby and Ainsworth is meant to apply to the entire complicated chain of processes resulting in infantile bonding, while the four learning theories considered are only appropriate for applying to each separate link of this chain. Therefore, these learning theories cannot possibly be alternatives to "the" ethological theory; and in considering separate links of the entire process, we must realize that one particular learning theory may best fit the data for one link while an alternative theory seems to hold for another.

Ethologically-oriented research has led to the notion that infantile attachment is brought about by guided development or genetically-programmed learning. Through evolution a gene pool has become established which ensures that the anatomy as well as the behaviour of individuals of a species will function sufficiently well to give the species a fair chance to survive and reproduce in its ecological niche. The way in which different genes manage to express themselves in structural and behavioural features varies. In the development of behaviour, genes may use learning processes to a greater or lesser extent, or the formation of motor patterns as well as releasing mechanisms (typical

sensory sensitivities). This notion has emerged from Lorenz's concept of "imprinting." Until now the principle of guided development has been most extensively worked out for the morphogenesis of bird song. The evidence so far collected on this topic shows that a relatively small amount of genetic information is extended stepwise through learning processes partly superimposed on one another. Via such processes, unlearned preset parameters lead to learned ones, which further guide the development of behaviour. The control by the genes is sufficient to secure a high probability that the biologically necessary knowledge will be acquired, and that the adoption of maladaptive and dangerous habits or responses will be prevented, provided that the animal grows up under normal conditions – i.e. the conditions that originally (i.e. in evolution) determined the characteristics of the genotype. One cannot expect the genes to provide protection against the numerous odd experimental situations that researchers are able to conceive. Consequently, I am not so surprised that it is possible to create attachments experimentally "to things that bear very little resemblance to any biological being." Also, the fact that the young have no protection against maltreatment by a partner is not surprising. Such a protection is built into the genes controlling the behaviour of the parent, but if that protection fails (in most cases probably through some disturbance in a program of guided development of the parent concerned), what else could the altricial young do but stick to the "parental object"? In most animals, strong mechanisms (the function and evolution of which would deserve much more study) are present to prevent the adoption of offspring of conspecifics; the young animal has no other place to go.

Research on the comparative psychology of learning has yielded considerable information concerning factors that can influence learning processes. To me it seems likely that of the various learning mechanisms postulated, more than one are likely to really exist. However, experiments have usually been carried out in environments and on problems that were limited in variation and evidently more inspired by human situations than by the original natural condition of the experimental animals. In my opinion there is an urgent need to investigate, with the ideas and methods of comparative psychology in mind, the factors determining learning processes under natural conditions, and for a great many different instances in the same animal species. I would therefore plead for a procedure in which the various theories and hypotheses would be tested for their value as building elements in the organized procedure of the development of complicated species-typical behaviour. I would advocate use of the concept of programmed

development as a backbone for these attempts.

Such investigations should start from an unbiased collection of data, followed by experiments based on deductions therefrom. They should not be primarily focused on attempts to invalidate either theory A or theory B. Theories, models, and hypotheses are rarely entirely wrong. They are always temporary stages that need to be tested, corrected, and amended. This is, for instance, what has happened with Lorenz's imprinting concept. Although I fully agree that Lorenz made premature generalizations that later had to be taken less dogmatically and amended, one should not write off his ideas, as Rajecki et al. do, by simply saying that his postulates were incorrect. Disposing of theories in that way amounts to throwing out the baby with the bathwater. Fortunately this has not happened in the case of imprinting, considering the vast amount of important ongoing research based on imprinting concepts.

by Dalbir Bindra

Department of Psychology, McGill University, Montreal, P. Q., Canada H3A 1B1

What next? A perceptual-motivational approach to attachment

In all the telling criticism, defensive clarifications, eclectic proclamations, and useful suggestions made in the thoughtful treatment of infantile attachment by Rajecki et al. (1978) and numerous commentators, one general point seems to have been overlooked. The failure of the various theoretical frameworks within which the phenomena of attachment are currently being approached is not surprising. For the most part those frameworks give inadequate explanatory accounts of adaptive behavior of *any* kind; their inadequacy in dealing with attachment merely reflects their failure in general. Thus their painstaking review of the phenomena of attachment could have been used by Rajecki et al. as an occasion for indicating the precise conceptual defects that make the various instinct, learning, and motivational frameworks inadequate, and outlining what might be the elements of any revised or new theoretical framework for dealing with the phenomena of attachment – and adaptive behavior generally. It is a pity that we do not have the benefit of their judgment on this point to guide further work.

Among the erroneous assumptions made by several current formulations is one about the relation between motivation and action. It is assumed, often implicitly, that the occurrence of any action, whether foraging, copulating, nursing the offspring, attacking, or retreating, depends on a single, unique motivational state. In fact, there is no one-to-one relation between a certain unique process – an instinct or a motivational state – and a single particular type of action. For example, eating, on the part of a hungry animal, depends not only on food-related motivation, but also on how familiar the animal is with the situation, what other opportunities or dangers lie therein, what effort is required to reach and ingest the food, and so on. Similarly, the pattern of responses displayed by a hungry mother wolf when an intruder arrives at her lair must arise from a mixture of hunger, attachment to her litter, and hostility evoked by the intruder. All actions, including the responses that define attachment, are outcomes of a multiplicity of motivational states, each of which is under constant modulation by internal organismic conditions and environmental incentive stimuli (appetitive and aversive), which may be primary (unconditioned) or secondary (conditioned). This point has been documented and elaborated elsewhere (Bindra 1974; 1976 cf. Balles 1972; Estes 1972).

A second erroneous assumption common to almost all current approaches is that representations or templates of different types of responses, once developed through maturation and learning, lie ready-made in the brain and are then *activated as such* (by a combination of motivational states and eliciting stimuli). This assumption of the activation of preexisting responses makes it impossible to develop plausible accounts of the variety, flexibility, and spontaneity of responses that emerge in relation to any given attachment object – or any other specified biologically significant (incentive) stimulus. An alternative view of response production is that any adaptive response is constructed afresh each time it occurs, and that its construction (that is, its form and aim) is guided by the spatiotemporal layout of unconditioned or

conditioned incentive stimuli, given the prevailing organismic state of the animal. From this viewpoint, the response that emerges in relation to a given attachment object at a given time is not a replication of some preexisting, inherent or acquired, "attachment response," but is a resultant of the spatial layout and temporal imminence of the incentive and nonincentive stimuli that define the situation. It is known that the response of a rat to a certain novel object depends on whether the object is presented in a familiar situation or a strange one (Sheldon 1969); Sluckin's commentary (*BBS* 1978, 3:458-59) mentions that expected attachment responses may fail to occur when aspects of the test situation are altered. Some of the principles that determine what exact response will emerge with a given spatial distribution and temporal order of significant stimuli in the situation has been outlined in fuller accounts of a perceptual-motivational framework of response production (Bindra 1976; 1978).

The question of how the particular responses, grouped for convenience as "attachment behavior" (e.g., approach, maintained contact, secure-base effect, separation distress, maltreatment effects), are produced at particular times is obviously one that requires detailed analysis. The point of this commentary is that such analyses may be better guided by the perceptual-motivational view of response production than by the older views of instinct, learning, and motivation. Responses in relation to an attachment object that appear perplexing from the viewpoint of other theoretical schemes may be quite readily understandable if approached within a framework that incorporates the ideas of motivational multiplicity and fresh response construction in the determination of adaptive behavior of all kinds (Bindra 1976; 1978). Within such a framework, it would not be inexplicable that a young boy, walking while holding his mother's hand, might tighten his grasp when a stranger looks at him, but a moment later may let go of the mother's hand and dash across the street to see a monkey or a snake. Nor would it be perplexing to see a child, who has been spanked by its mother to the point of distress, approach and cling to the mother when she also happens to be the most appetitive object in the situation at that time. Nor would it be difficult, within the perceptual-motivational framework, to specify the conditions under which an habitual attachment response may fail to occur.

by Marc H. Bornstein

Department of Psychology, Princeton University, Princeton, N. J. 08544

Two questions for a general theory of infantile attachment

Rajecki et al. (1978) provide a brief but sophisticated evaluation of the comparative literature on infantile attachment. Their goal is to assess the concordance of data and theory pertinent to a nodal problem in developmental psychology. Two issues relevant to the assumptions and expectations of their treatment need to be discussed. They are reviewed here in a question-and-answer format.

Question 1. What proof have we that there exists a social bond particular to infancy? The behaviorists' legacy is that psychologists must mistrust "black-box" theorizing. Human and comparative developmentalists typically define social bonds operationally and in terms of observables (e.g. "signals," Bowlby 1969; "approach," Schneirla 1965; or "duration in proximity," Gewirtz 1972). Reliance on one or another specific dependent measure may narrowly confine the generalizability of an investigator's results, and even when investigators adopt a strategy wherein they focus on the functional equivalence of a variety of observable behaviors that "facilitate maintenance of adult-infant proximity," problems still arise. A principal and inescapable assumption of this research is that infantile attachment is accurately reflected in the eye or grid or stop-watch of the beholder. Protestations to the contrary, attachment is still a hypothetical derivative or black-box state only presumed to reflect itself in quantifiable behaviors. The potential psychological distance between behavior and state is therefore crucial to the success of a theory of attachment. In these authors' own words, "the sheer *quantity* of the reaction is less impressive to us than the *quality* of the reaction," and such reactions, they admit, are "usually unspecified." This being so, a schism between data and theory is hardly surprising.

What is beclouded in the attachment literature is that the simple aggregation of "functionally-equivalent" observables does not compensate for this underlying deficiency. There is really no reason to believe, therefore, that signaling, approach, or proximity conclusively manifest an experiential or motivational state of "attachment," since 1) each observable may have a cause independent of some internal state of attachment; 2) signalling, etc. may genuinely reflect attachment in one organism but not in another; 3) signaling, etc. may genuinely reflect attachment at one developmental stage but not at another; and 4) attachment may genuinely occur in the absence of signaling, etc. The situation may be likened to one that dogs population research – inferences about a psychological state (e.g. crowding) are frequently unwarranted and are not equivalent to manipulation of a correlated physical variable (e.g. density). Drawing a distinction between state and behavior would seem theoretically imperative and might permeate this otherwise excellent literature review as much as the qualitative – quantitative distinction does. How developmentalists (or other psychologists) face this challenge is open. Answers to the first question on the meaning of what is measured relate intimately to answers to the second question.

Question 2. Why should there ever be a general theory of attachment? Attachment is a "good problem," and one could make a strong *prima facie* case (as Rajecki et al. do implicitly) that it is treated optimally in a comparative framework. Despite this effort, there is no reason to believe that comparative data and theory in attachment need necessarily converge into a single "General Theory." Indeed, the specific comparisons made in this review, as well as general considerations, suggest the opposite.

First, a minor point: There is no strict phyletic comparison here, since birds, dogs, and men have evolved in different lines of specialization. Hodos and Campbell (1969) have pointed succinctly to the problems of comparability and prediction created by inappropriate interspecies comparisons. Although a phylogenetic comparison was not the author's intent, the different evolutionary histories of these species force the reader to question a central assumption of Rajecki et al.'s review: namely that manifestations of attachment (e.g., maltreatment, secure base effects, and protest) are comparable beyond analogy. Attachment in species who inhabit different ecological niches may be functionally homoplastic but need not be superficially so. It may therefore be unreasonable to develop a theory of attachment with such variegated sampling. (Rajecki and Lamb (1978) deal with this point adequately in their response.)

Can a single view encompass a phenomenon so complex as attachment? At minimum, and in the absence of any data, attachment immediately implies the integration of perceptual, cognitive, and social functions to which evolutionary history, ontogenetic stage, and environmental experience are all inextricably pertinent. Rajecki et al.'s review forces us to account for the features of attachment eclectically, as we do presently for gender identity, morality, or other psychological phenomena. There is no consensual view; learning theory, ethology, or other approaches cannot by themselves explain the range of phenomena associated with attachment. One wonders why it should be otherwise. The functions of theory are multiple: to focus thinking and to generate crucial research are as important as comprehensive explanation.

But these few comments are only prolegomena to any general theory involving psychological explanation.

by John Bowlby

The Tavistock Clinic, London N.W.3. 5BA

The Bowlby–Ainsworth attachment theory

Following my friend, Mary Salter Ainsworth, I wish to congratulate Rajecki and his colleagues for a careful review (Rajecki et al. 1978) which in almost all respects does justice to the ethological theory of attachment that we have advanced.

Most of such shortcomings that Rajecki et al.'s review have already have been commented on, but I would like to return to two. The first concerns the authors' suggestion (on page 432) that human infants show "rapid shifts" in the figure towards whom their attachment

responses are directed. This is simply not true, and it is certainly not supported by the findings of the foster-care study undertaken by Robertson and Robertson (1971), to which Rajecki et al. refer. In the first place, in this study each of the four children was thoroughly familiarized with the foster parents and their home during a period of not less than four weeks prior to being taken into care; in the second, there was evidence throughout the children's stays that, though they treated their foster mother as an attachment figure, none of them was content with the arrangement. The two older ones, aged 2.4 and 2.5, in particular, distinguished sharply between the foster mother and their missing mother. Not only did each of them express longing for mother, but, when mother returned, each went straight to her and distanced themselves from the foster mother. The behaviour of the two younger children, aged 1.5 and 1.1, although less clear-cut, was full of conflict. Thus it is no more true to suggest that these children showed a rapid shift in their attachment figure than it would be to say the same because a child is willing to be left for an evening with a familiar babysitter.

In regard to attachment figures, infants show a clear hierarchy of preferences, especially when they are distressed: first, a preferred figure; next, various familiar substitutes; finally, and only when no one else is available, perhaps a kindly stranger.

The fact that an infant may appear reasonably content with a figure other than his preferred one can often be misleading. A striking example from another species is provided by Reite et al. (1978) in a study of pigtail macaques. When the mother of an infant is removed from a group, it is most unusual for another female to foster the infant; the latter habitually becomes distressed and shows characteristic changes in body temperature, heart rate, and sleep patterns. In one experiment, however, another female took pity on the infant, aged about twenty weeks, and acted as his foster mother. The infant responded and appeared content with the arrangement. The physiological measures told a different story, however. Even though the infant was not obviously distressed, the physiological changes were typical of a separated infant.

My second point concerns the protective function of attachment behaviour and the difficulty Rajecki et al. discuss concerning the fact that in an atypical environment, behaviour may be directed towards objects that afford no protection. Yet I imagine they would hardly raise this sort of objection in regard to the reproductive function of sexual behaviour, because this behaviour may sometimes be directed towards objects with which breeding is not possible.

Although outside the scope of the original article, perhaps I could say a word about the clinical implications of these different theories, since I believe that which of them is adopted has far-reaching consequences for practice.

When attachment to a preferred person by an infant, a child, or an adult is viewed ethologically, the ensuing behaviour is likely to be respected as being as intrinsic to human nature, as are, say, sexual behaviour or eating. As a result, when a clinician meets with an individual who is responding to separation or loss with various mixtures of protest, anger, anxiety or despair, he is likely to recognize the behaviour as being made up of the natural, even if perhaps inconvenient, responses that any human being would be expected to show in the type of situation that the individual concerned is in. The clinician is likely, then, to take relevant action of whatever kind lies within his power.

When, however, attachment behaviour and responses to separation and loss are viewed in terms of one or another of the learning theories, among which the one favoured by Anna Freud (1946; 1954) has (despite the contrary statement by Rajecki et al., page 418) been immensely influential in clinical circles, the scene looks very different. When a baby has been recently fed, crying is seen as merely demanding attention, and picking him up, as likely to result in his crying more. When a child is protesting loss of his principal attachment figure and is demanding her return, he is regarded as having been spoiled. An adolescent or adult who is apprehensive, lest he be deserted by his attachment figure, is regarded as overdependent, hysterical, or phobic. The clinician's actions are likely, then, to be as disapproving as they are irrelevant. Considerations of these kinds were in fact the ones

that led me in the early fifties to question the relevance to clinical problems of any form of learning theory [see Eysenck: "The Conditioning Model of Neurosis" *BBS* 2(2) 1979] and to adopt an ethological approach as the one that seemed best fitted to helping me understand the clinical phenomena with which I was daily faced in my work with children and parents. Advanced, as Mary Ainsworth remarks, as an open-ended theory, intended to help understand the clinical and research findings then current, the fact that it has also stimulated a great deal of further research, both clinical and experimental, gives me the utmost satisfaction.

by Gary Greenberg

Department of Psychology, Wichita State University, Wichita, Kans. 67208

Approach/withdrawal theory and infantile social bonding

Some years ago Sigmund Koch (1969) cautioned us that unless our data-collection efforts slowed and were accompanied by data syntheses, we would be bogged down helplessly and hopelessly in a meaningless sargasso sea of data. Until very recently this is where we were with imprinting or social-bond phenomena: much data, little synthesis. In fact, P. P. G. Bateson (1971) once remarked of the state of imprinting that the very many experiments performed have served to render the phenomenon ill-defined. It seems quite clear that comparative psychology must eventually sift through the vast mountains of its existing data to finally achieve one of its major goals: the discovery and elucidation of general laws of behavior.

The utility of this sort of activity is illustrated rather nicely in the recent target article by Rajecki, Lamb, and Obmascher (1978), which has provided us with a much needed theoretical overview of infantile social bonding. To be sure, even a casual search of the vast imprinting literature shows the phenomenon to be defined as a following response, approach response, filial response, species recognition response, and so on; further, these responses are typically studied in diverse ways – following, approaching, with runways and circular alleys, as two (and more) choice discriminations, with operant manipulations, and so on. Clearly we are not dealing with a simple, clearly-defined behavior, but rather with a complex *set* of behaviors collectively labeled "infant social bonding" (imprinting).

It is not surprising, then, to learn from Rajecki et al. that there are some "eleven different perspectives on the formation or dynamics of attachment in infants." As the authors point out, there is often substantial overlap among these different perspectives, sometimes to the extent that theoretical differences seem to vanish. One wonders, for example, about the utility of identifying both a classical conditioning and an operant conditioning model rather than subsuming their respective facts under a single rubric such as "learning theory."

But the real point of this paper is the following: Rajecki et al. have unfortunately glossed over one of the potentially more significant theoretical positions in contemporary psychology by rejecting Schneirla's epigenetic approach to behavior because, in their words, it "could not claim substantial support in the domains of imprinting." Let me attempt to outline the important aspects of Schneirla's theory and show how it may in fact be adequate to account for imprinting phenomena and very much more.

I think it may be appropriate to consider Schneirla's approach to the development of behavior an "epigenetic orientation," although some would disagree with the use of this term. Nevertheless, if only for argument's sake, I feel comfortable in identifying Schneirla as an epigenetic theorist. I might add that Gilbert Gottlieb (1978) seems to agree with this, and he has pointed out that while it may be impossible not to adopt an epigenetic orientation in comparative psychology, there may in fact be several epigenetic points of view. Epigenetic thinking, therefore, is not necessarily monistic.

As you might expect, then, according to Schneirla, behavior is a developmental phenomenon or process, appearing and changing as a result of all the experiences encountered by a maturing organism from the moment of its conception to the time of its death. Two fundamental concepts central to this thinking are that of *maturation*, which refers to the organism's "growth and differentiation together with all of their

influences upon development (p. 269)." (This quote as well as the two following ones are from a collection of Schneirla's writings edited by Aronson, Tobach, Rosenblatt, and Lehrman 1972), and *experience*, "which connotes *all* stimulative influences upon the organism through its life history, (p. 269)." Thus, organisms develop, and in the course of their development they interact with stimuli. This interaction shapes and directs their behavior.

One of the key concepts in Schneirla's system involves the biphasic processes underlying approach and withdrawal responses by organisms to stimuli. In Schneirla's words, "for all organisms in early ontogenetic stage, low intensities of stimulation tend to evoke approach reactions, high intensities-withdrawal reactions with reference to the source (p. 299)." While there is certainly a great deal more to Schneirla's theorizing, including a use of the concept of integrative levels with respect to an organism's behavioral plasticity and its evolutionary status, it is sufficient for our purposes to focus on the simplicity involved in 1) maturing organisms, 2) experiencing stimulation, and 3) early in their lives approaching or withdrawing from that stimulation as a function of its intensity. The system is so parsimonious that one wonders why it has not become more widespread in contemporary psychology.

The adequacy of this theoretical position has been substantially documented in the writings of Schneirla, and I refer those interested to his works, notably: *Principles of animal psychology* (with Norman Maier 1935 and 1964) and the more recent collection of papers, *Selected writings of T. C. Schneirla* (Aronson et al. 1972). In addition, a number of related works have been directed toward demonstrating the adequacy and utility of Schneirla's position, among them: *The development and evolution of behavior* (Aronson et al. 1970), and *The biopsychology of development* (Tobach et al. 1971).

The important point, again, is that the theory is quite simple and parsimonious, enabling us to derive explanations where previously only suggestions were available. Thus, one can account for Tinbergen's hawk-goose phenomenon rather nicely by appealing to approach/withdrawal (A/W) theory rather than to some innate conceptual scheme present in young birds. In the hawk configuration the model enters the visual field fully and abruptly and may be considered a rather intense visual experience, thus eliciting withdrawal responses. In the goose configuration the model enters the visual field much more gradually and thus may be considered a much less intense visual stimulus, thereby eliciting approach – no instinct, no unexplained phenomena, just a testable proposal subject to falsification; all the prerequisites of a true scientific explanation.

Experimental verification of the A/W concepts has been spelled out in Schneirla's works. More recently Gerald Turkewitz and his students and collaborators have provided many human examples of its validity. For example, in infants (as the theory suggests) approach movements (finger extension) were found to be elicited by weak visual stimuli, and withdrawal movements (finger flexion) to be elicited by more intense stimuli (McGuire and Turkewitz 1978). Earlier, Hammer and Turkewitz (1975) found a similar pattern of responding with directional eye turns elicited by an auditory stimulus; infants turned their eyes towards an effectively weak auditory stimulus and away from an effectively intense one. Another study investigated the effectiveness of stimulus intensity in infants' visual attention from a developmental perspective (Ruff and Turkewitz 1975), finding that "infants younger than 9 to 10 weeks attend to visual stimulation on a different basis than older infants (p. 708)." In summary, then, there appears to be support for Schneirla's A/W concepts, and these are central to his system as a whole.

The real question of concern, however, is the utility of the theory for infantile social bonding or imprinting. In his early work on imprinting Moltz said: "During the past five years we have worked with over 600 birds and I cannot think of a single instance in which approach was initiated as the [imprinting] object moved towards the bird; indeed, when traveling in that direction, the object will occasionally evoke withdrawal responses." (1963, p. 126)

The suggestion is being made here that approaching and retreating visual stimuli are differentially intense. An object that retreats produces a retinal "image" of decreasing size and, hence, stimulates fewer and fewer retinal elements; one that approaches, however, subtends a

progressively greater visual angle and, hence, stimulates more and more retinal elements. Thus, a retreating object is the less intense one and in Schneirla's scheme would elicit approach responses; an approaching object is more intense and would elicit withdrawal responses.

While it still remains to adequately test this notion empirically, it is inappropriate to reject it, as Rajecki et al. have done, without further study. There are, in fact, some tantalizing reports that offer important support for the A/W postulations.

Tronick (1966) showed that chicks would imprint, by approaching, to a radially decreasing shadow – i.e. the shadow's retinal projection got progressively smaller. While he did not compare the effectiveness of this stimulus with one that gradually increased in area, he did show it to be as effective as a real receding object; these results are clearly predicted by the A/W scheme, although in Tronick's analysis they were contrary to A/W expectations. Tronick suggested that a shadow decreasing in area results in a progressively brighter and hence more intense stimulus.

That there are two ways of looking at the intensity of one stimulus seems to indicate that "stimulus intensity" is not an easily determined parameter and may be a much more complex concept than previously imagined. Indeed, Turkewitz addressed this point in his analysis of the adequacy of A/W theory in human infant behavior.

In addition, research by Bengtson (1974) makes this point abundantly clear: he says that "A large object is not always a high-intensity stimulus." Bengtson showed stimulus size and intensity to be related to changes in angular speed and distance from the eye. To assume, then as many imprinting researchers have done, that size is the crucial aspect of stimulus intensity is to make an erroneous assumption.

Finally, Metcalfe's (1974) research on the influence of prehatching visual stimulation in imprinting behavior illustrates the complex relation between maturation status and reactivity to stimulation. One may recall that maturational status is one of the important considerations in Schneirla's theoretical scheme. Perhaps, then, intensity is a *relative* characteristic of a stimulus, its degree being influenced not merely by its physical characteristics, but by the relative maturational status of the organism as well. What may be intense for one organism may be weak for another one, and even for the same organism at another point in time.

These arguments speak to the necessity of a more adequate definition of the stimuli we use in imprinting studies. We naively assume that "the" stimulus we present to an organism is "the same stimulus" that that organism responds to.

In summary, then, it would seem that the rejection of Schneirla's theoretical position is premature and uncalled for. It is not that the various other theories presented by Rajecki et al. are wrong; it is just that they are incomplete and deal only with isolated segments of an organism's behavior or ontogenetic history. For example, learning-theory approaches to social bonding represent only one feature of the dynamic process, *but surely not all*. In the opinion of Schneirla and the other epigenetic theorists, there is much more to behavior – any behavior – than learning, ethology, and opponent processes. And we are merely on the threshold of developing an adequate methodological orientation that will enable us to determine the origins of behavior.

by Douglas A. Kramer and William T. McKinney, Jr.

Department of Psychiatry, University of Wisconsin, Madison, Wisc. 53792

Ethology: the natural model

Rajecki et al. (1978) have written a comprehensive paper, which we believe is important both because it reviews the ethological literature on attachment and also discusses two clinically important issues in psychiatry: separation and maltreatment.

We believe that an ethological model is particularly useful in describing naturally occurring behavior (Kramer and McKinney 1979). Models are only approximations to reality, and when used to describe broad categories of naturally occurring behavior such as attachment, they should be relatively inclusive – that is, account for all of the available data. In this case the model has been useful in understanding attachment behavior. However, it might also be used to describe other

forms of naturally occurring behavior such as communication, aggression, and migration [see Eibl-Eibesfeldt: "Human Ethology" *BBS* 2(1) 1979].

It is our opinion that models based on experimental conditions are not necessarily incorrect, or without utilitarian value, but that they are, by definition, less likely to approximate in their design the conditions of evolutionary adaptedness. Therefore they are less likely to account for naturally occurring behavior than a natural model. In many respects the most valuable experiments are the natural ones. The best laboratory is the natural environment over an evolutionary time span, and the most useful energy behind an experiment is that of natural selection. The ingenious work of Tinbergen (1958, pp. 193–199) in sorting out the color patterns and range of two larval forms of a butterfly species as related to predatory factors is an example of the use of this sort of natural experiment. Another is Hailman's (1965) investigation of optical signals in swallow-tailed gulls in the Galapagos, which is demonstrative of the way in which a thorough understanding of an evolutionary niche can lead to an understanding of species-specific behavior patterns.

However, even though the ethological model may be the logical and appropriate one for naturally occurring behavior, laboratory-based models are also useful. We differ somewhat from Rajecki et al. regarding the nature of the potential usefulness of these laboratory-based models. In our view, they are most appropriately used to understand specific mechanisms of a given behavioral system, rather than the system as a whole. Thus, knowledge of imprinting phenomena, classical conditioning, or operative learning theory, as derived for a given species, can be helpful in understanding the mechanisms involved in the development of attachment systems. Such knowledge can also be useful in understanding variations from individual to individual, whether induced experimentally in a laboratory setting or nonexperimentally in a natural setting.

A laboratory model not mentioned in the review and suggested to us by Hailman (1977), which may describe aspects of the imprinting process, and therefore the attachment process as well, is the reinforcement model of Premack (1959). Perhaps there is a natural analog to his laboratory model whereby contingencies are imposed upon freely-occurring behavioral events, and thereafter the probability of the contingent behavior reflects the probability of the noncontingent behavior irrespective of reinforcement properties of the former. For instance, looking at the mother may be a relatively high probability event compared to going toward mother; however, for all practical purposes, the latter event is contingent upon the former. Therefore, an aspect of attachment behavior (increased proximity to the object) may be, in part, related to the probability of looking at the mother and not particularly to how reinforcing she is.

The section on maltreatment effects is a stimulating discussion of an apparent anomaly in the attachment system. We believe the term "maltreatment" is unsuitable for the category of behavior to which it is applied, as it is value-laden and potentially difficult to operationalize. Maltreatment is just as much a function of the behavior being described as it is of the context in which it occurred. Bowlby (1969, pp. 226–227) quotes three references to field observations of attacks by dominant males upon juveniles. The effect was analogous to that observed in the laboratory by Harlow (1961), Rosenblum and Harlow (1963), and Seay, Alexander, and Harlow (1964) – that is, that attachment behavior was elicited. In the field, under natural circumstances, it was observed that the context of this behavior was one in which the dominant male had sensed a predator or other danger, and the result (the elicitation of attachment behavior) was of positive survival value.

The maltreatment effects discussed in Rajecki et al.'s review may not be paradoxical, given hypotheses that could be derived from consideration of ethological field studies. Natural ecological conditions may have been more conducive to attachment behavior in these maltreatment situations than what today, in a laboratory or a civilized community, may seem logical. With regard to another "maltreatment" model, an interesting aspect of the research with the "motherless mothers" reported by Harlow, Harlow, Dodsworth, and Arling (1966) was the behavior of these mothers in subsequent pregnancies. The abusive treatment that the first infants of these mothers received did not occur in subsequent pregnancies, and in fact eight out of nine

subsequent pregnancies were accompanied by mothering that was at least adequate. Therefore, this experience, albeit in the laboratory, apparently without practice effects for prosocial attachment behavior, and despite being characterized by interactions that were injurious and even lethal to the infants, had an effect on the subsequent development of these "motherless mothers" ultimately resulting in a biologically significant improvement in their mothering behavior.

Physiological components of attachment behavior have also been studied. Klopfer (1971) demonstrated that attachment in goats was dependent upon internal changes in the does at the time of delivery. Mothers accepted alien young and displayed separation distress upon their removal if they were presented within five minutes of the mothers' parturition experience. Klopfer speculated that these internal changes were mediated by a hypothalamic hormone, such as oxytocin, released in association with dilatation of the cervix at delivery. Furthermore, he showed that olfactory factors were operative in the discrimination of one's own young from alien young. Mothers could attach to their own young for up to several hours if olfaction was not experimentally interfered with by cocaineization of the nasal epithelium.

We have a position that differs from that of the authors with regard to species and individual specificity in response to separation experiences. Such specificity is not a contradiction to an ethological approach to attachment behavior, but rather an acknowledgement that all behavioral systems are species-specific (including the degree of individual variation possible within a given behavioral system). A species-specific behavior pattern is always an evolutionary compromise between competing selective factors, and it can only be understood in light of the environmental niche in which it originally developed. We feel that these deviations between species in response to particular situations, such as separation, are opportunities to refine our theory through a better understanding of the origins of apparently diverse behavior.

We found Rajecki et al.'s target article a comprehensive and stimulating paper, for the reasons described above. We disagree only with certain emphases, as they have been articulated. The paper's greatest value may be the discussion of apparent maltreatment and the attachment system.

by Michael Lewis

Educational Testing Service, Princeton, N. J. 08541

Cognitive factors in attachment

The article by Rajecki et al. provides a rather important review of several of the theories pertaining to the development of attachment(s). While the authors do not make a point of attempting to define exactly what is meant by attachment, I think it safe to assume that they would agree that it contains at least six features, these including:

1. an affectionate quality of the relationship – perhaps most closely akin to what we human primates call love (see, for example, Ainsworth 1969, p. 1015);
2. proximity-seeking and maintaining behaviors – varying with the age of the organism and proximal as in touch, or distal as in looking or vocalizing (see, for example, Lewis and Ban 1971);
3. an enduring quality – thus, the feeling state is not transitory but lasting, exactly how long is not clear;
4. the attachment relationship(s) are unique and are differentially expressed with respect to the attached as opposed to other relationships. Thus, we may consider them to be the difference between love relationships and others that are not love relationships;
5. attachment is an all or none phenomenon; that is, all children become attached to their parents. What is commonly studied is not whether the infant is attached but only the value of the attachment;
6. attachment is a construct referring to a tie or bond, implying that there is no set of "specific behaviors which mediate the attachment, and indicate its presence" (Lamb 1974, p. 382). These features have been specified more precisely in Weinraub, Brooks, and Lewis (1977).

While there is general agreement that these features are relevant to the discussion of attachment, there is one aspect of this relationship that receives almost no consideration; this is the role of the child in the attachment relationship. Rajecki et al. do raise this issue, albeit briefly.

To us, this aspect of attachment theory has received the least attention, since most theories have stressed the role of the adult (usually the mother). What is needed is a broadening of the concept of the relationship between child and other to include the cognition, volition, or plans of the child. For example, one stays close to another for many reasons: the salience of the stimulus, as in imprinting, its history of reinforcement properties, its past associations, and so forth. There are, however, other reasons, having to do with what the child knows about the caregiver. The child's cognitive ability, the ability to conceptualize and understand, must be included. Because both birds, for example, and human children seek proximity does not imply that the same process is at work. The same type of behavior can be under the service of very different sets of motives, unlearned and reflexive in the former case and learned and cognitive in the latter.

It is of interest to note that attachment appears, or is said to appear, in humans at the same time when a variety of significant cognitive milestones are reached; for example, object and person permanence, self recognition, the ability to separately compare two objects spatially located in different places, and so on. The rise of these cognitive abilities at the same time that attachment is displayed suggests important relationships between them both.

In order to explore this relationship, a study of two-year-old children's response to separation has been undertaken (Weinraub and Lewis 1977). Maternal departure and separation was studied, since conceptually these represent different tasks for the child. A conceptual model of the determinants of separation distress was postulated in which three variables were thought to affect the child's distress: the child's cognitive abilities, maternal departure styles, and the amount of mother-child interaction preceding departure. It was predicted that the more advanced the child's cognitive abilities, the better able the child would be to form a cognitive structure to interpret the separation event. After all, the child has to interpret the mother's behavior, especially as seen in the attachment paradigm, where the mother is appearing and disappearing in a strange environment every three minutes! Mother's departure style has not been studied before, since mothers have been instructed as to how to depart and what to say. In the present study she could say what she pleased, and three styles were identified: sneak out, instruct as to what is to happen ("I'm leaving, will be back soon"), and instruct and tell the child what to do when she is gone. These styles were predicted to affect the separation distress by influencing the child's cognitive structure of the separation event. Finally, the mother-child proximity or affective relationship was measured by the amount of mother-child interaction. Our results clearly indicate that *both* the amount of mother-child interaction and the cognitive influences affect the child's response to separation.

This study, along with others (Shain 1976; Lester, Kotelchuck, Spelke, Sellers, and Klein 1974), have implicated the child's cognitive ability in the attachment relationship. Any theory attempting to consider attachment across a wide phylogenetic scale has to be prepared to consider 1) the child's role in the attachment relationship, and 2) the child's (and mother's) cognitive structures and the role they play. Moreover, any theory of attachment seeking to consider parents and children across a wide phylogenetic range also has to postulate different mechanisms or processes for the same outcome. It is not necessary to consider that the same process is involved in the attachment of a young bird and child. Toward such a differentiated theory, we would propose that the role of cognition is of prime importance.

by Mary Main

Department of Psychology, University of California, Berkeley, Calif. 94720

The ultimate causation of some infant attachment phenomena: further answers, further phenomena, and further questions

Rajecki et al.'s (1978) is a well organized review of attachment theories and phenomena – one which should engender research as well as controversy. Its chief fault seems to me to lie in an insufficient attention to, or understanding of, evolutionary or "ultimate" causation. In this commentary I shall first show that an attention to evolutionary causa-

tion can help to explain some of the phenomena that the authors profess to find puzzling. In addition, I shall call attention to a major attachment phenomenon that the authors have essentially omitted from consideration: this is *avoidance* of the attachment figure under stress conditions, and it is not uncommon.

The authors of this review call for a more complete understanding of "attachment phenomena." Let me begin, therefore, by noting that for an ethologist (e.g. Tinbergen 1963), four specific kinds of questions regarding a pattern of behavior must be answered before we can claim understanding. These are, first, the two (related) questions of *development* and *causation*: these are sometimes called the "proximate" questions, or questions of "mechanism." In the case of attachment behavior this amounts to the difference between asking "How did this individual come to show this preference for this particular conspecific?" and "What made him show attachment behavior toward it at this particular moment?". The second set of closely related questions are the questions of *survival value* and of *phylogeny*, also called questions of ultimate causation. "What is attachment behavior good for – how does it affect survival and reproduction?" "What selection pressures account for the differences between species – for example, in the mechanisms of mother-identification?" It is now a truism that efforts to solve any and all of these questions should be preceded by a careful description of the development, form, and occurrence of the behavior.

If these are indeed the requirements for a full understanding of this behavior pattern, then it must be admitted that our present understanding is discouragingly limited. Among the six major "theorists" cited by Rajecki et al., only Bowlby and Ainsworth have provided a description of attachment behavior occurring outside the laboratory; Bowlby alone has been concerned with survival value; and not one of the major theorists has made extensive phylogenetic comparisons.

In their review of attachment theories the authors have given a major (perhaps central) place to "ethological" theory, and yet only a few lines are concerned with explicating evolutionary issues and interpretations. This is surprising, since that feature which makes a study "ethological" is not primarily its strong descriptive base but rather its tie to behavioral biology and its concern with questions of ultimate causation. In what follows I shall concern myself primarily with this level of explanation.

The violence of the response to separation and the "secure base" phenomenon are, indeed, two of the central phenomena of human infant attachment. In the protected environments in which we now live, neither of these behavior patterns seem rational – e.g., an infant left for a moment in a strange toy-filled room is in no danger. It was Bowlby's genius to suggest that these phenomena could not in fact be completely explained on the level of "proximate" causation. He suggested that attachment behavior evolved to serve the biological function of protection from predation.

It seems to me, however, that the full importance of the attachment behavioral system in infancy cannot be understood when it is conceived merely as one behavioral system among others serving one among many possible functions. In choosing to emphasize the predator-protection function of attachment behavior, (1) Bowlby may have weakened our opportunity for appreciating its full importance. Protection is only one of the primary survival problems: others are, e.g., the gaining of food (2) and of shelter (3). As infants, some animals are dependent on the proximity of conspecifics for solving only one of these problems. We and our closest primate relatives are, however, dependent on adults for all of these and others (See Konner 1977). The maintenance of proximity to attachment figures simultaneously serves several biological functions and is the *sine qua non* for infant survival.

These facts should solve the problem of understanding the violent "insecurity" that human infants show in response to brief separations: on the ultimate level their feelings of insecurity refer to a real, if historical, danger. In addition, it explains the "secure base" phenomenon, which the authors have dealt with only superficially. Thus, although referring to "ethological" theory, the authors refer only to proximate mechanisms or even correlates (the fact that set-goals may shift) to explain secure-base phenomena. This is an outstanding

omission. The fact that maintenance of proximity to an attachment figure has so many survival functions means that, in the behavioral hierarchies of infants, maintenance of proximity must come first. While the alternative to exploration at a given moment could be less familiarity with some class of objects in later life – and this certainly in the long run affects survival and reproductive success – the alternative to maintenance of proximity to the attachment figure could be a fairly immediate death. Exploration can then be afforded in the presence of the attachment figure, but the infant whose mother moves off must devote itself to finding her. (The authors err, incidentally, in contrasting an "actual" motivation to explore with one that is dis-inhibited (p. 431); there need be no real difference.)

Let us now consider the ways in which an attention to evolution and natural selection can enhance our understanding of bond formation. The authors of this review profess puzzlement that "infants of many species form bonds to objects that are not typical of any species' environment, or even to objects that are sources of maltreatment." Somewhat to my amazement, they imply that they find these facts *contradictory* to "the (ethological) notion that infants are predisposed to being influenced by the expectable features of the social environment." In fact, these inappropriate attachments provide the very evidence that supports this ethological notion.

For each species of infant, natural selection needs only to develop a means of identifying the prospective attachment figure which, in the *expectable* environment, is *sufficient* to distinguish her from others. For some precocial birds this means the first thing seen moving; for nonhuman primate infants, the first thing they can cling to. The criteria are not infinitely broad (objects that cannot be comfortably clung to are not taken as rhesus attachment figures), but they are efficient, simple, and work to select a biologically appropriate object in the expectable environment. Criteria sufficient to identify the mother in the natural environment, however, confuse the laboratory animal – in the laboratory, cloth may be clung to and foam rubber may be followed. There is *no* reason why identification mechanisms should be identical in different species.

In the human case, what is the identification mechanism? Those proposed are essentially salience (Cairns), service (Gewirtz), and social interaction (Ainsworth and Bowlby). The authors have unfortunately overlooked this emphasis upon *social* interaction. Simultaneously they have failed to present the evidence (Ainsworth 1963; 1967; Schaffer and Emerson 1964) that human infants form attachments to adults who engage in *social* interaction and games – and this in preference, if necessary, to those who merely engage in caregiving interactions.

BBS readers may be interested in a more specific proposal regarding mechanism, put forth by John Watson (1972). In a series of studies on the early learning of response contingencies (learning in infants approximately two to four months old) Watson and his colleagues constructed crib mobiles, which either did or did not turn in response to infant head or foot action. The initial aim was simply to discover how quickly infants could discover the association. Over a series of days, however, the infants who had contingently-responding mobiles began to behave towards them as they normally would to "social" stimulation – i.e., they responded to the mobile's "response" with vigorous smiling and cooing. Using this and other evidence, Watson has proposed that the perception of a contingent temporal relation between infant activity and immediate subsequent stimulation may be the major initial influence guiding the human infant to classify conspecifics as "social objects." This perception should always occur when people play social games (interact socially) with young infants, and since infants of this age are usually incapable of having any significant effect on the inanimate environment, it is probably the *only* circumstance in which such contingencies have occurred in (precontingent-mobile) species history. We may therefore go even further and note that the human infant is unlikely to err in picking attachment figures if it favors those who have played social games with it; these are most likely to be the individuals who take the greatest interest in the infant and are most likely to provide protection. Thus this mechanism, which can be tested, ties in an altogether reasonable way to human infant survival. It would be interesting to see whether monkey infants would be interested in

such things (they should not be). At the same time, these considerations *might* provide some understanding of "blanket-attachment." Blankets provide contact-comfort; we are primates; and this fact alone could make them a candidate for a quasi-attachment. The Watson hypothesis, however, renders them even likelier candidates. For human infants, confined to cribs, blankets may be the single inanimate object most likely to consistently respond contingently to infant activity. If this is true, these inappropriate attachments may provide not a contradiction but a proof that "infants are predisposed to be influenced by the expectable features of the social environment."

Turn now to the problem of "maltreatment effects." Neither Cairns's nor Watson's proposals are inconsistent with the possibility of the formation of bonds under conditions of sheer maltreatment. Note, however, that the authors have not presented a single such case. In every study discussed in which bonds to maltreating objects were formed, these objects additionally provided the conditions (e.g. contact comfort) required for the formation of attachment. ("Non-zero social orientation" in three out of six severely punished dogs is not, of course, attachment.) The implication that bonds may be formed to "anything at all" is unfortunate; monkeys seemingly do not bond to cold, wire mothers, and it would be surprising if they formed bonds to cold, maltreating, wire mothers.

The formation of bonds to maltreating figures is not surprising if we suppose that biologically the infant knows – perhaps, in some sense, knows above all else – that this figure is the single means to survival. Even a malicious, malignant, maltreating, attachment figure has demonstrably not yet killed the infant. What would be surprising would be a *given* infant's *preference* for a maltreating object over one that is *not* maltreating; *that alone* could be considered maladaptive. In the only such tests conducted (Barett 1972), infants preferred the nonmaltreating object.

Peculiar maltreatment effects – that is, the irrational return of the abused to the abusing object – were first noted by Darwin (1972) in his voyage to the Galapagos; they are presented along with an explanation of the mechanism. He physically assaulted a Galapagos sea-lizard, as he stood on a promontory, and each time tossed it seaward. Although "possessed of perfect powers to swim away" from him, it returned each time to the point on which he stood. "Perhaps this singular piece of apparent stupidity may be accounted for by the circumstance that this reptile has no enemy whatever on shore, whereas at sea it must often fall prey to the numerous sharks. Hence, probably, urged by a fixed and hereditary instinct that the shore is its place of safety, whatever the emergency may be, it there takes refuge." This is an "ultimate" account made sheerly at the level of mechanism, and it is essentially identical to Bowlby's.

Severely maltreating mothers cannot be common in any species, and perhaps this account of "maltreatment effects" on the proximate level is sufficient. On the other hand, it is at least conceivable that some biologically-based strategy has been developed to deal with maltreating mothers. Trivers (1974) has pointed out that some parent-child conflict is to be expected, given the fact that parent and offspring are not genetically identical, and he points to weaning conflicts as an example. During the period of weaning in infant rhesus, maternal rejections grow increasingly sharp, and up to a point they are met with *increasing* infant insistence upon proximity (i.e. "maltreatment effects"), including "tantrums." Trivers interprets these responses not as the fall-out from proximate mechanisms, but as a deliberate offspring strategy that deceives the parent regarding how much more investment the infant needs before being independently viable. This claim would be theoretically untestable, except that Trivers proposes that at some point the infant will *cease* its relentless returning to the rejecting (and increasingly exhausted) mother, because it sees that it is costing itself more in terms of half-copies of itself in its young siblings than it is gaining on its own. If differing terminal points for the seeming order to "increase-proximity-in-response-to-rejection" could actually be predicted and substantiated in specific cases, we might indeed begin to see in some manifestations of this pattern an infant's active evolutionary strategy. Alternatively, the infant may be engaging in an ultimately passive response to mechanisms which have in themselves no particular survival function.

Some intriguing proposals have also been made for evolutionary causes for infant *separation* phenomena. First, Kaufman and Rosenblum (1967) have proposed that the period of despair or depression, in which the infant has a very low activity level, has the biological function of conserving energy for an infant whose searches for the mother have proven not only futile but exhausting. Second, Charlesworth¹ has suggested that this low activity may serve as an anti-predator device by helping to camouflage the infant. Third, Trivers (1974) has offered explanations in terms of "ultimate" causation for changes in mother-infant interaction following separation. These are again based on data collected in experiments with rhesus (Hinde and Spencer-Booth 1971). Following reunion with its mother, the rhesus infant spends more time on her than it did before separation, although, since it is aging, it ought to be spending less. Trivers finds these data to be "consistent with the assumption that the infant has been selected to interpret its mother's disappearance as an event whose recurrence the infant can help to prevent by devoting more of its energies to staying close to the mother." This is in fact not far from the explanation put forth by Bowlby (1973).

A still more interesting phenomenon, which Trivers attempts to explain in terms of evolutionary strategy, is the effect of the pre-separation mother-offspring relationship on the offspring's behavior upon reunion. The more frequently a rhesus infant was rejected prior to separation, the more distress it shows upon reunion and the greater its role in maintaining proximity to the mother. According to Trivers, "These data support the assumption that the infant interprets its mother's disappearance in relation to her pre-departure behavior in a logical way: the offspring should assume that a rejecting mother who temporarily disappears needs more offspring surveillance and intervention than does a nonrejecting mother who temporarily disappears."

Fortunately or unfortunately for the development of overriding theories, some human infants respond to separation and reunion with a pattern of behavior that has not been found in experiments conducted with other species: they avoid the attachment figure, looking away, turning away, and actively ignoring efforts made by the attachment figure to attract their attention. In major separations (separations of two or more weeks) this behavior appears following the stages of "protest" and "despair" already described by Rajcecki et al. – i.e. following stages in which anger is openly expressed toward the (visiting) parent. Avoidance on reunion is associated with a return to normalcy *in the separation environment* (the authors are misleading in stating that "normalcy" follows "despair," for this reason); but, in that proximity-avoiding is replacing proximity-seeking in the parent-child relationship, it represents something far from normalcy in terms of attachment. It is, nonetheless, an attachment phenomenon: it is specifically attachment figures who are treated in this way, while others may be greeted actively. Although this behavior may be confined to human infants, I believe the phenomenon is as important as the other three described by the authors.

Separation phenomena cannot be explained solely in terms of attachment, because other behavioral systems (aggression) and special states (depression) after arise, and these have no necessary relation to attachment. (For this reason I think it is unreasonable to ask a theory of *attachment* to specify all the behaviors that can be observed on separation.) None of the theories described in the article are capable of predicting this phenomenon, and that is not surprising since they are aimed at "predicting" the appearance of attachment behavior in circumstances in which it has generally made its appearance. Here is an antithetical behavior pattern, avoidance, appearing in those same circumstances.

When toddlers undergo major separations in which no consistent caregiver is available to substitute for the original one, data collected to date suggest that almost all children respond to reunion with avoidance. If, on the other hand, such a person is available during a separation of the same length, the attachment figure is not avoided (Robertson and Robertson 1971). This shows, of course, that the behavior cannot be accounted for solely by separation but must be a function of some process not inevitably connected. Heinicke and Westheimer (1965) have suggested that the responsible process is a

rise in anger toward the parent. Robertson and Robertson suggest that this is very appreciably lessened when a consistent caregiver is available to function as a subsidiary attachment figure.

A large number of home-reared one-year-olds (infants who have never experienced major separations) have now been studied in a miniature separation situation conducted in the laboratory; here the parent twice briefly leaves and returns to the infant (Ainsworth et al. 1978). The majority of United States infants respond like primate infants everywhere – with distress to separation, and with active efforts to achieve and maintain contact with the mother on reunion. Angry behavior, whether expressed in angry screaming or in tantrum movements, is not uncommon. Two thirds of infants who do not avoid show anger.

Some infants who have never experienced major separations nonetheless avoid the mother on reunion in this laboratory situation. About half of the United States infants seen to date at least look away from the returning mother once, as she appears during one of the two reunion episodes, and about a fifth avoid and ignore her throughout the episode. Infants who strongly avoid the mother attend to toys or other objects throughout much of the separation. If they show distress, they are easily settled by a stranger. Avoidance of the mother within a given reunion episode is negatively related to openly angry behavior (Main²);

Fortunately, it has been possible to combine these laboratory-stress situations with home observations (Ainsworth et al. 1978), or with observations of play involving both infant and parent (Main²). There are no indications that infants who avoid the mother under stress are infants who have failed to become attached to her; indeed, some show strong distress on separation from her in the home situation. Avoidance does not appear to reflect a “type” of infant. Infants who will later avoid show no early failures to cuddle, and, while avoidance to a given parent is stable, infants may avoid one parent in one laboratory observation while in a separate session clinging to another (Main and Weston³). Data concerning father-infant relationships are not yet fully analyzed, but the following are findings from three mother-infant samples. First, the more episodes of violently angry behavior toward the mother occur in the home and play situation, the more avoidance of the mother occurs in the stress situation. Second, avoidance of the mother is strongly associated with her physically-expressed rejection of the infant in the home and play observations (Main²; 1977; in press). Third, we have found a similar syndrome of avoidance and angry behavior directed toward adult caregivers in what we believe is the first controlled investigation of battered (abused) infants. (George and Main, in press). Finally, avoidance of the parent on reunion in daycare centers is strongly correlated with avoidance of the same parent in the laboratory situation (Blanchard and Main, in press).

These phenomena raise several problems for investigators concerned with attachment. On levels of both proximate and ultimate causation we may ask why this active avoidance appears at all, precisely in situations in which it is least expected. It is especially puzzling in the light of the “ultimate” explanation put forth by Trivers.

On the proximate level I have pointed to the theoretically unresolvable conflict situation in which a physically rejected, attached infant finds itself (Main 1977). In terms of immediate causation I have suggested that avoidance appears in rejected infants in stress situations as an alternative to angry behavior – that, in ethological terms, it functions as a kind of “cut-off” (Chance 1962; Eibl-Eibesfeldt, BBS 1:4, 1978) to rising aggressive tendencies when these conflict with affiliative tendencies.

The survival value of attachment behavior is so clear that it is difficult even to begin to ask questions regarding the “ultimate” causation of a pattern of behavior that is – on first appearances – antithetical. It may well be that avoidance has no function of its own and no association with maternal rejection other than whatever is ancillary to proximate mechanisms. The latter is almost certainly true if we assume that mothers never even moderately rejected their young infants in the environment in which present patterns evolved. This is assumed in the statement that “Infants whose attachments are insecure will behave somewhat maladaptively within their environment . . . and would be less likely to survive in the long run.” This implies, note, that what an infant does when it is rejected (“insecure”) is to behave in a way that

still *further* lessens its *already* reduced chances for survival. To me this seems unlikely. If true, at what point on the continuum of maternal rejection does this positive feedback for increased mortality begin? If true, maternal rejection could only appear in the most protected of environments.

Until very recently most of us may well have conceived of major behavior patterns as having evolved to deal with the vicissitudes of the environment, with the establishment and maintenance of simple relationships to conspecifics, and not at all with vicissitudes in these latter relationships (see G. C. Williams 1966). The evolution of the attachment-behavioral system is certainly conceived of in this simple fashion.

It is possible, however, that there are conditional behavioral strategies that may still function within the bounds imposed by the “primary” system – in this case maintenance of proximity to the mother – and that may be activated only when special interpersonal conditions warrant. In this case, avoidance of the attachment figure may function as a conditional strategy, which paradoxically permits whatever proximity is possible under conditions of maternal rejection. The infant, rather than reaching a high pitch of angry behavior and distress, avoids, maintains control, and continues exploration. This behavior (in contrast to the alternatives) may either simply fail to encourage increased mother-infant distance in stress situations, or it may actually elicit caregiving (Lamprecht⁴).

As views, it seems to me that both possibilities are acceptable and await a reasoned disconfirmation; as stances, neither the “adaptive” nor the “maladaptive” views of avoidance is pleasing. If we take the “maladaptive” view too easily, it seems to me that we are falling into the clutches of an unjustified mental-health psychology, which is pleased to find evolution as well as personal experience condemning insecurity, rejection, and deviance. If we take the “adaptive” view too easily, we find answers everywhere, and this should be anathema to scientists. The latter point is unfortunately nicely illustrated within the confines of this review; Trivers (1974) and I have each found it possible to present “ultimate” accounts for the reunion behavior of rejected infants in two closely-related species, when in fact, upon reunion, rejected infants in the two species, seem to behave precisely antithetically. This may be because the pre-separation rejection behavior of rhesus mothers does not really resemble that of human mothers, because the mothers behave differently upon reunion, or for some other reason. However, the rhesus infant that has undergone long-term separations from its mother apparently approaches rather than avoids her on reunion, whereas even “accepted” human infants respond to major separations with avoidance. The species differences here are most intriguing.

This commentary has been chiefly intended to alert those interested in attachment to the meaning of considerations of ultimate causation, to the intriguing possibilities to which this in turn leads, and to the need for explaining yet another attachment phenomenon (avoidance), if only a human one. Although the consideration of ultimate causation may irritate some readers from the start, I think it should be, at least in principle, pleasing to those who desire to consider the infant as taking an active role in its own development. It may do this on the proximate level, but it may also do this on the ultimate one.

NOTES

1. Charlesworth, W. (1978) Personal communication.
2. Main, M. (ms. in prep.) *Avoidance of the attachment figure in infancy*.
3. Main, M., and Weston, D. (ms. in prep.) *There is no relationship between the quality of infant-mother and infant-father relationships*.
4. Lamprecht, J. (1978) Personal communication.

by Marsha Weinraub

Department of Psychology, Temple University, Philadelphia, Penna. 19122

Separation distress in human infants: A multifaceted, multidetermined response

Rajcecki et al. (1978) take a refreshing approach to attachment. While some recent investigators have focused on describing qualitative differences in the security of caregiver-infant relationships and others

have focused on describing the infant's social behaviors within a social network, these authors have considered attachment as a theoretical issue. They avoid conceptual and operational definitions of attachment – all of which could be easily challenged – and examine the success of theoretical accounts of social bonding to account for specific phenomena within the attachment area. Their emphasis on attachment behaviors as phenomena to be explained by attachment theory holds promise for a better understanding of parent-infant relationships.

However, if we are to successfully compare and refine models, the factors affecting attachment behaviors must be made explicit. To this end I would like to choose one phenomenon singled out by the authors – separation responses in human infants – and demonstrate the complexities involved in describing and explaining this behavior.

Separation distress is a multifaceted response. Not every involuntary separation provokes distress; the occurrence and intensity of separation distress depends not only on the state of the organism but also on the characteristics of the situation, the nature of the observed response, and the age of the organism.

Nature of the situation. Separation distress is more frequent in unfamiliar than in familiar settings (e.g. Ross et al. 1975). More significantly, different factors may influence the expression of separation distress in familiar and unfamiliar places. Those infants who show separation distress at home tend to be those who appear otherwise insecurely attached (Ainsworth, Stayton, and Bell 1974). The mother's departure style – whether she leaves through a familiar or unfamiliar door (Littenburg et al. 1971) or what she says on departure (Weinraub and Lewis 1977) – also affect infants' separation response. The number of toys in the room, the novelty of the toys, and the presence of a stranger all mediate the effects of separation (Cortez 1972). Finally, the extent of parent-infant interaction immediately prior to separation in an unfamiliar place predicts separation distress (Weinraub and Lewis 1977). These factors demonstrate the importance of stimulus properties of the separation situation and underplay the role of the infant's relation to the absent object.

Nature of the response. Responses to maternal departure should be distinguished from responses to maternal absence. Little relation between these two responses has been observed, and the evidence so far suggests that these responses may have differential determinants (Weinraub and Lewis 1977; Smolak, Weinraub, and Feureman, in preparation). Departure responses may be related to sophisticated cognitive awareness of impending absence and a desire to influence the mother, while separation responses may indicate unsophisticated cognitive skills and an inability to cope with a novel situation.

Even during the maternal absence situation, a variety of responses have been observed. Distress behaviors observed immediately after departure range from disoriented emotional distress (random activity, crying, screaming and kicking), to protest and anger (demands for the mother), to search behaviors (movement to the door), to general sadness (apathetic, listless wandering and inability to play). These behaviors change over time within the situation and with increasing separation experiences.

Considering each of these responses as interchangeable forms of distress leads to confusion, especially since different models focus on different behaviors. Cairns's model predicts behavior indicative of disorientation and disruption following initial departures; Gewirtz's model predicts emotional distress early in the infant's separation history, giving way to protest patterns dependent on the infant's reinforcement history; and ethological theorists predict protest and search behaviors.

Age of the organism. Although attachment continues and may even grow stronger between the ages of one and three years, separation distress follows a different course. Beginning near the end of the first year, it increases to a peak at around 18 months and then declines, so that by two years, only about one in two infants are visibly distressed on brief separations.

The meaning of separation distress may change with age. Its onset may indicate the formation of a special attachment relationship, while

its offset may indicate that the infant has developed coping strategies to deal with his emotional distress. At different ages, different factors predict separation responses. For example, Gallas and Weinraub (1978) observed that for 18-month-olds, talking to mother in a free-play situation prior to departure resulted in more distress during separation, but for 24-month-olds, talking predicted *less* distress. At 18 months, separation distress may indicate a sophisticated social response, an awareness that an important person is unavailable. By 24 months, separation distress may be a relatively unsophisticated response, indicating that the child is unable to rely on language and internal representations to cope with the object's absence. These data suggest that researchers should be very cautious in looking for an explanation of separation distress that will hold across age.

Separation distress as a multidetermined response. Separation distress also has varied determinants. Even focusing on the same behavior within the same situation, multiple factors may contribute simultaneously to individual differences. Weinraub and Lewis observed that two different factors simultaneously predicted individual differences in 2-year-olds' play during maternal absence: characteristics of the interaction preceding departure and the extent to which the child was cognitively prepared to interpret the separation event. Qualitative aspects of the mother-infant relationship may be important insofar as they affect either the amount of interaction prior to separation or the child's cognitive sophistication. Previous experiences with separation and reinforcement history may also shape children's separation responses (Tulkin 1973; Smolak et al. in prep.).

Conclusion. Assessments of the power of different explanations of separation distress will be necessarily superficial until the complexity of the behavior is taken into account. Because separation distress is a multifaceted response, attempts to describe it must 1) consider specific features of the separation setting, 2) specify the behaviors used to index distress, and 3) be sensitive to differences in the meaning of separation distress at different developmental levels. Because separation distress is a multidetermined event, more than one model may be needed to explain this behavior.

Contrary to Rajecki et al.'s implication, we *do not know* what the phenomenon of attachment looks like. However, we *can* paint a fairly detailed picture of one particular attachment behavior: separation distress. Studies specifically designed to understand the multiple determinants of separation distress may provide more useful information regarding the origins of social bonding than studies that have used separation distress as an attachment index.

Research should be directed toward illuminating the effects of quality and quantity of interaction between partners, cognitive skills, and previous separation experiences on separation responses. Wherever possible, experimental manipulations should be used. Cross-age as well as cross-species comparisons should be considered. Similarities in 1) patterns of responses, 2) determinants of individual differences, and 3) mechanisms of successful coping in response to the loss of loved ones during different life stages may yield information not only about the development of separation distress, but also about the origins of human personality.

EDITORIAL NOTE

*Asterisk indicates author has already replied to commentary in Response accompanying target article (Rajecki et al. 1978).

Authors' Response

by D. W. Rajecki and Michael E. Lamb

Department of Psychology, University of Northern Iowa, Cedar Falls, Iowa 50613;

Center for Human Growth and Development, University of Michigan, Ann Arbor,

Mich. 48109

Infant attachment: some final thoughts about theory and method

For the most part, the above commentaries address general issues in the infant attachment area rather than the specific

conclusions reached in our review. The net effect is extremely constructive, and in our reply we wish only to amplify some of the points raised. To this end, one can divide the commentaries into two categories, one focused on theoretical and one on methodological issues.

Some thoughts on theory. Several of the commentators – notably, BAERENDS, BINDRA, GREENBERG, KRAMER & MCKINNEY, and MAIN – suggest novel conceptualizations and approaches in response to certain problems highlighted in our review. Clearly, our critical appraisal of infantile attachment theory and research has not led researchers to abandon the complicated issues concerned; instead, it has prompted valuable clarification and refinement. We could not have hoped for a more gratifying effect. Of some significance is the observation that these commentators have not built with equal frequency upon the various perspectives we evaluated. It was our original conclusion that the ethological approach was the most serviceable, whereas the learning approaches were the weakest. If one reflects on the commentaries – both the earlier ones and those printed above – this conclusion is strongly supported. The current emphasis in the field is one that seeks to integrate the psychology of the child with the biology of the species.

Some thoughts on methodology. As noted earlier, several of the additional commentaries addressed methodological rather than theoretical issues. BOWLBY, LEWIS, MAIN, and WEINRAUB all suggest renewed interest in separation and reunion behavior. Interest in response to separation was originally stimulated (at least in part) by its clinical implications (see Bowlby). However, the continuing study of involuntary separation effects may yield insights into the attachment process. Consider, in this regard, Main's discussion of "identification mechanisms" and Weinraub's research on the antecedents of variations in separation protest. As Main points out, identification mechanisms may differ from species to species, and the study of separation effects may provide a way of discovering these different mechanisms. In a recent illustrative study, Rajecki et al. (in press) recorded the behavior of chicks in the presence of their imprinting objects prior to an involuntary separation at about one week of age. A large number of factors were studied – including the effects of the visual and auditory characteristics of the objects, the manner in which the objects interacted with the birds (ranging from gently to abusively), frequency of vocalization in the presence of the objects, and proximity to (but not contact with) the objects – none of which predicted the amount of distress at separation. Interestingly, the amount of physical contact shortly before separation was positively correlated with the amount of distress. These results were obtained in two independent experiments.

Some might be tempted to suggest that WEINRAUB's findings concerning the frequency of maternal vocalization and Rajecki's findings concerning contact in chicks imply cross-species parallels. Others – mindful of previous claims regarding the importance of auditory and visual cues in birds (Immelmann 1972; Gottlieb 1976) and tactile cues in primates (Harlow 1958) – might acknowledge only that the studies point toward additional species-specific identification mechanisms. Whereas the amount of interaction may govern the intensity of separation protest in children, touching may govern the intensity of protest in chicks.

However, this leads into controversial issues that have dogged research on human infant attachment for years. The interrelated issues concern the meaningfulness of the notion of "strength of attachment" and the distinction between attachment behaviors and attachment bonds. Ainsworth has long held that there is little to be gained from attempts to

assess individual differences in the "strength" of attachment. To be sure, it appears possible to rank the preferences of a given infant (e.g., Lamb 1977a; 1977b), but there is such great individual and contextual variability in the occurrence of the index behaviors that it appears meaningless to compare the strengths of different infant's attachments. Ainsworth and her students (Ainsworth et al. 1978) have been especially noted for demonstrating that much of this variability is predictable; attachment behavior is intensified, for example, when infants are tired or frightened. The fact that the intensity of attachment behavior varies depending on the internal psychological state of the infant lies at the base of another claim – that there is no one-to-one correspondence between the attachment behaviors and the underlying emotional bonds they appear to mediate. (Around these issues, students of socio-emotional development and comparative psychology may well part company.)

Instead of investigating strength of attachment, therefore, Ainsworth has pioneered research on individual differences in the patterning or organization of attachment behaviors. She has shown that the way infants behave toward attachment figures when distressed is related to the way the adults have behaved toward the infants in the preceding months. This has provoked interest in the precise mechanisms whereby variations in the behavior of attachment figures are translated into enduring and formatively significant orientations in young infants. For example, MAIN has paid special attention to the origins and significance of avoidant behavior, whereas one of us (Lamb, in press) has taken a different tack, suggesting that all the behavioral patterns (including avoidance) reflect the infant's expectations regarding parental behavior. It is probable that these issues will dominate research on human infant attachment in the years ahead. As Main demonstrates, ethological considerations remain extremely valuable in the formulation of hypotheses in this area.

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